Why Internet Crimes Against Children And Retrospective Case Analysis Matters

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Abstract

This report makes the case for why Internet Crimes Against Children (ICAC) deserves attention from researchers, engineers, and policymakers — and why cross-case retrospective analysis in particular represents an underinvested frontier. It is structured as a five-part argument: why the internet's compressed thirty-year evolution from research network to mass social infrastructure has created conditions that bad actors exploit in ways that outpace protective response; what the current response ecosystem looks like across law enforcement, policy, technology, and research; what measurable changes technology has produced over the past decade; where the genuine limits of current enforcement and detection approaches lie; and what retrospective case analysis and systems like CaseLinker can contribute. This is a position piece grounded in data. The goal is not to overstate what automated tools can accomplish, but to articulate clearly why the analytical gap at the case level matters — and why it should be filled.

Contents

1. Why the Internet, and Why Children

The World Wide Web became publicly accessible in August 1991. Commercial internet service providers reached mass-market consumers by 1994–1995. The first recognizable social networking platforms appeared around 2003–2004. Mass smartphone adoption — which moved internet access from a desk to a pocket, from optional to universal — accelerated after 2010. Instagram launched in 2010. Snapchat in 2011. Discord in 2015. TikTok reached Western markets in 2018. In thirty years, a communication technology went from a research curiosity to the primary infrastructure of social, economic, and civic life for billions of people.

That is an extraordinarily compressed timeline for a transformation of this magnitude. And it carries a specific implication: the population most immersed in digital life today — children and adolescents — is navigating an environment whose risks, norms, and exploitation vectors are still being discovered in real time. The adults responsible for their safety, the institutions designed to protect them, and the legal frameworks governing their online conduct were all built for a different world.

This gap between the speed of technological adoption and the pace of protective response is the central problem this report addresses.

1.1. A Technology That Is Genuinely New — and Double-Edged

The internet's benefits for children are real and well-documented: access to educational resources, social connection across geography, creative expression, and community for young people who might otherwise be isolated. These are not incidental features. They are among the reasons the internet achieved such rapid and deep adoption among young populations.

But every structural property that makes the internet valuable also makes it exploitable. Reach — the ability to communicate with anyone, anywhere — removes the geographic constraints that historically limited predatory contact. Anonymity — the ability to present any identity — enables deception at a scale impossible in person-to-person interaction. Storage — the ability to store, copy, and distribute digital content indefinitely — transforms individual acts of exploitation into explicit, distributable material. Speed — the real-time nature of digital communication — compresses the grooming process and enables rapid escalation.

> *The internet did not invent the exploitation of children. It removed the geographic, logistical, and social constraints that historically bounded it — providing unprecedented reach, anonymity, and distribution infrastructure to anyone willing to misuse it.*

These properties are not bugs introduced by bad actors. They are features of an advancing technological world, designed for legitimate purposes, that bad actors can and have learned to exploit systematically. This is the core challenge: you cannot eliminate the exploitable properties without removing the technology that makes the internet valuable. Every policy and technical response to child exploitation online is therefore a negotiation with tradeoffs, not a clean fix.

1.2. The Compounding Effect of Advancing Technology

Risk changes as the capabilities available to bad actors advances. This is not a theoretical concern — it is a pattern already visible in the data.

End-to-end encryption, introduced to protect the privacy of billions of legitimate users, simultaneously allows communication and distribution of explicit material and CSAM between perpetrators. Child Sexual Abuse Material (CSAM) — the legally and clinically preferred term over "child pornography," which implies consent where none can exist — refers to any visual depiction of the sexual exploitation or abuse of a minor, criminalized under 18 U.S.C. §§ 2252–2256 at the federal level and equivalent statutes across all fifty states; explicit material refers more broadly to sexually graphic content that may not meet the legal threshold for CSAM but is nonetheless used in grooming, coercion, and exploitation.

Generative AI, developed to assist with creative and productive work, can been weaponized to produce synthetic child sexual abuse material at a scale that overwhelms — NCMEC documented a 1,325% increase in AI-generated CSAM reports in a single year between 2023 and 2024 [NCMEC, 2024]. Open-source voice and image synthesis tools, released to democratize content creation, enable perpetrators to fabricate convincing impersonations for social engineering at minimal cost. Cryptocurrency networks, designed for financial privacy, provide payment infrastructure for exploitation networks that is substantially harder to trace than conventional financial channels. Almost every young individual on the planet has a high powered device with a camera in their pocket that connections them to billions of other people.

The pattern is consistent: each wave of technological advancement that reaches mass adoption is typically followed by its incorporation into exploitation methodologies. The same digital infrastructure that accelerates this adoption also enables bad actors to find one another — forming communities where exploitation methodologies, evasion techniques, and detection-avoidance strategies are actively shared and distributed. The interval between technological release and criminal adoption also appears to be shrinking. What took organized networks years to operationalize in the early 2000s can now be replicated by individual actors with consumer hardware and freely available software. As the digital landscape evolves, so too does the collective knowledge of those who misuse it.

> *As the capability ceiling for legitimate users rises, the capability ceiling for bad actors rises with it. Anonymity tools become more effective. Distribution infrastructure becomes more resilient. Synthetic content becomes harder to distinguish from authentic material. Any framework for child protection that does not account for this trajectory is already falling behind*

This dynamic has a specific implication for how protective technology must be built. Static detection systems — those calibrated to the threat landscape at a fixed point in time — will eventually fall behind. Hash-matching databases require ongoing updates. Behavioral detection models require retraining as grooming language evolves. Legal frameworks require amendment as new crime typologies emerge. Addressing this issue means understanding the ever evolving landscape of the computing field and its potential impacts for exploitation.

1.3. The Exploitation of Children Is Not New

Child sexual exploitation is among the oldest documented human crimes. It predates the internet by millennia. The children targeted have always been vulnerable by virtue of age, developmental stage, and dependency on adults — not because of anything the internet created. What changes with technology is not the underlying predatory dynamic but the scale, the speed, the geographical reach, and the evidentiary complexity of cases.

Understanding this matters for anyone who works in this space. The problem is not a technology problem — technology is a vector. The problem is a human one, and any response has to be human-first. Technical tools can reduce exposure, improve detection, and accelerate investigation. They generally cannot preemptively address the conditions that produce exploitation, and deployment of such preemptive methods come with privacy and ethical concerns. Overstating what automated systems accomplish is not a benign error: it produces misallocated resources, misplaced confidence, and policy frameworks that address the surface level vectors of harm without addressing its roots.

1.4. The Research Gap This Work Addresses

Reading public ICAC case reports — published annually by task forces like the Arizona ICAC — reveals something that statistics alone do not: the texture of individual investigations. The platforms that appear. The perpetrator behaviors that recur. The investigative approaches that are used and how they vary across jurisdictions and years. The nature of the these crimes.

What those reports also reveal is a structural analytical problem. Cases contain patterns that are visible even to a non-expert reading through a small collection of reports manually. But there is no tool that makes those patterns visible at scale. Reading forty-seven case summaries manually is possible. Reading forty-seven thousand is not. If the goal is to understand the landscape of child exploitation in the United States as it has actually manifested in investigated cases over the last decade — not as reported by platforms, not as estimated by surveys, but as documented by law enforcement — there is currently no systematic way to do that.

That gap is why I designed CaseLinker, to aid in understanding this issue space. The remainder of this report documents what the existing response ecosystem looks like, what it has and has not accomplished, and why the analytical layer that CaseLinker represents is a meaningful and necessary addition to it.

2. Current Approaches Across Law Enforcement, Policy, Technology, and Research

There is a diverse, talented, and extremely motivated ecosystem of practioners working to address this issue. However, the response to internet crimes against children is distributed across multiple sectors, each doing meaningful work and each operating largely independently of the others. Understanding what exists — and where the gaps are — requires looking at all enforcement efforts and how they integrate in this ecosystem.

2.1. Law Enforcement: The ICAC Architecture

The primary U.S. law enforcement response to internet crimes against children is organized around the Internet Crimes Against Children Task Force Program, a federally coordinated network of 61 task forces representing over 5,400 federal, state, and local agencies [OJJDP, 2024]. The program was established in 1998 — the same year the NCMEC CyberTipline was created — and has grown steadily since, expanding jurisdiction, staffing, and technical capacity over nearly three decades.

In fiscal year 2024, ICAC task forces conducted approximately 203,467 investigations and arrested more than 12,600 offenders [OJJDP, 2024]. For context, in FY 2021, the number was approximately 10,400 arrests from 137,000 investigations [U.S. COPS Office, 2023]. The growth in investigative volume is substantial, and would grow as digital media proliferates across the internet.

FY 2024 ICAC Program Snapshot [OJJDP, 2024]

- ≈203,467 investigations conducted
- >12,600 arrests of suspected offenders
- ≈46,000 law enforcement professionals trained
- $39.9 million in program funding
- 61 task forces, ∼5,500 affiliated agencies nationwide

Investigations are classified as proactive (investigators initiate contact in undercover capacity) or reactive (a tip or complaint triggers the investigation). Both types generate substantial caseloads. The FBI's Child Sexual Exploitation Unit, Homeland Security Investigations (HSI), and the Internet Watch Foundation's international counterparts operate in parallel, handling cases that span jurisdictions too broad for state-level task forces.

The challenge ICAC task forces face is not primarily operational competence. It is scale. The ratio of CyberTip reports to available investigators is staggering. In 2024, NCMEC received approximately 20.5 million reports of suspected child sexual exploitation — equivalent to 29.2 million separate incidents when adjusted for bundled submissions [NCMEC, 2024]. Against that volume, 12,600 arrests represents successful case resolution for a fraction of a fraction of the reported volume.

This is not a failure of law enforcement. It is a resource and infrastructure problem.

2.2. Policy: A Reactive Legislative Landscape

Federal policy in this area has been shaped primarily by three legislative frameworks: the PROTECT Our Children Act of 2008, which codified ICAC and established mandatory reporting requirements; 18 U.S.C. § 2258A, which requires electronic service providers to report apparent CSAM to the CyberTipline; and more recently, the REPORT Act of 2024, which expanded mandatory reporting to include online enticement and child sex trafficking for the first time [Congress, 2025].

The REPORT Act is significant. Its expansion of reporting obligations represents the most substantive statutory update to the reporting ecosystem in over a decade. But NCMEC's 2024 data tells a cautionary story: despite new mandatory reporting categories, overall CyberTip incident volume declined approximately 7 million from 2023 to 2024, partly due to declining platform compliance and partly due to the expansion of end-to-end encryption across major messaging platforms [NCMEC, 2024, Congress, 2025]. Legislation that requires reporting is only as effective as the infrastructure that enables platforms to detect the reportable activity.

At the state level, policy varies considerably. Prosecution outcomes, sentencing guidelines, digital evidence admissibility standards, and inter-agency data sharing protocols differ across jurisdictions in ways that complicate cross-case analysis and national pattern detection.

2.3. Technology: A Tale of Two Layers

The technology ecosystem for child protection operates on two distinct planes.

The first is *content detection* — the tools platforms use to identify known child sexual abuse material. PhotoDNA, developed by Microsoft in 2009 and used by major platforms Google, Twitter, Facebook, Reddit, Discord, and NCMEC, is the cornerstone technology. It creates perceptual hashes of images and compares them against a database of known CSAM, detecting matches even after modification. PhotoDNA has been described as highly effective in detecting known CSAM content, with low false positive and false negative rates [Steinebach, 2024, ARES, 2023]. Google's CSAI Match performs an equivalent function for video. Meta's open-sourced PDQ and TMK+PDQF algorithms enable smaller platforms to implement perceptual hashing independently. Thorn's Safer platform takes a complementary approach: where PhotoDNA matches against NCMEC's centralized hash database, Safer combines perceptual hashing against its own aggregated database of 82+ million verified CSAM hashes with an AI-based classifier layer designed to detect *novel* content that has never been hashed before — addressing one of hash-matching's core structural limitations [Thorn, 2023]. Safer also enables cross-platform hash sharing among its customers, allowing newly identified CSAM to propagate across the detection ecosystem without delay [Thorn, 2025]. According to the Tech Coalition's 2023 annual survey, 89% of member companies use at least one image hash-matcher, and 59% use video hash-matching [Tech Coalition, 2023].

The second plane is *case analysis* — the tools investigators use to understand the evidence they've collected. This layer is dominated by enterprise platforms: Nuix Neo for large-scale e-discovery and timeline reconstruction, Magnet Axiom for mobile and cloud forensics, Cellebrite UFED for device extraction. These platforms are powerful but constrained — expensive, device-centric, infrastructure-heavy, and rarely accessible to the small and mid-sized ICAC task forces that handle the majority of investigations.

Between these two planes is a gap: there are no widely deployed, accessible tools for case-level, cross-case narrative analysis. No system for asking, systematically, what patterns emerge when you look across hundreds or thousands of investigated cases. Analyzing how perpetrators behave across platforms and deriving prevention policies from recurring patterns remains difficult with current technologies.

2.4. Research: Meaningful but Incomplete

Academic research in this space has produced valuable tools and insights. Natural language processing work on grooming detection has developed models for identifying predatory communication patterns in chat logs [Kontostathis et al., 2010, Al-Nabki et al., 2023]. Computer vision research has advanced age estimation from images [Ricanek & Mahalingam, 2014]. Deep learning classifiers for CSAM detection have been evaluated on private and semi-public datasets [Laranjeira et al., 2022].

The UNICRI AI for Safer Children initiative, launched in 2022, has catalogued over 80 AI tools relevant to child protection investigations and trained over 1,000 investigators globally [UNICRI, 2022]. Thorn, a nonprofit technology organization, has built and deployed machine learning classifiers for platform-level CSAM detection and victim identification, reducing time-to-identification for victims from weeks to days in documented cases.

But research has a consistent limitation in this domain: it operates on file-level or content-level data. The case narrative — the full story of an investigation, its platform context, its perpetrator methodology, its relationship to similar investigations — remains largely unstudied at scale, because the data is difficult to study and the tools to analyze it does not exist publicly.

3. Direct, Measurable Changes Technology Has Made in the Last Decade

3.1. What Has Improved

Detection of known material at platform scale. PhotoDNA and hashing solutions have fundamentally changed the economics of distributing known CSAM on major platforms. In 2019, a dozen of the largest tech companies accounted for 99% of CSAM detection and reporting [Prostasia, 2020]. The hash-matching infrastructure currently deployed generates the overwhelming majority of CyberTip reports. Without it, the 20+ million annual reports would not exist — there would simply be no systematic detection at all. That is a real and substantial gain and one that demonstrates successful platform compliance and cooperation.

Victim identification. NCMEC's Child Victim Identification Program, launched in 2002, has identified over 19,100 children as of 2024 [NCMEC, 2024]. INTERPOL's International Child Sexual Exploitation (ICSE) database, which integrates PhotoDNA and supports global victim identification across more than 70 countries, had identified over 30,000 victims by 2022–2023 [MDPI, 2025]. These are not estimates; these are children removed from active abuse situations or whose perpetrators were identified and prosecuted.

ICAC investigative capacity. The ICAC program's investigative volume grew from approximately 11,206 arrests in FY 2010–2011 combined [OJJDP, 2012] to over 12,600 in FY 2024 alone [OJJDP, 2024] — alongside a dramatic increase in the complexity and volume of cases. Thousands of real children who were victims of abuse and neglect have been rescued. Training has scaled as well: ICAC task forces have delivered over 194,000 community outreach presentations since 1998 and trained 46,000 professionals in 2024 alone.

Platform coordination. The Tech Coalition and industry hash-sharing frameworks like Meta's Hasher-Matcher-Actioner (HMA) have enabled cross-platform signal sharing that would have been technically and legally complicated to operationalize a decade ago.

NCMEC CyberTipline reports went from approximately 100,000 in 2010 to over 36 million in 2023 — a scale increase that, while alarming in what it reflects about online exploitation, also represents an unprecedented improvement in detection and reporting infrastructure [UNICRI, 2022, NCMEC, 2024].

3.2. What Has Gotten Worse

AI-generated CSAM. The most significant emerging threat is the weaponization of generative AI for CSAM creation. NCMEC documented a 1,325% increase in CyberTip reports involving generative AI content between 2023 and 2024 — from 4,700 reports to 67,000 in a single year [NCMEC, 2024, eSafety Commissioner, 2026]. In the first half of 2025 alone, reports involving AI-generated exploitation jumped from 6,835 to 440,419 — a 64-fold increase in six months [HSToday, 2025]. The Internet Watch Foundation documented over 20,000 AI-generated images posted to a single dark web forum in a single month in 2023 [IWF, 2024]. Hash-matching is useless against AI-generated content: there is no prior hash to match.

End-to-end encryption. The expansion of E2EE across major messaging platforms has created significant detection blind spots. NCMEC has documented direct correlations between specific platforms' E2EE rollouts and declines in reported incidents. The 7-million-incident drop from 2023 to 2024 was attributed in part to E2EE expansion [Congress, 2025]. This is not an argument against encryption — the privacy arguments are real. It is a statement of the tradeoff.

Sextortion. Online enticement reports increased over 300% from 2021 to 2023 and a further 192% in 2024 alone, reaching 546,000 reports [NCMEC, 2024]. Financial sextortion — where offenders coerce money rather than seeking sexual gratification — has emerged as a distinct and rapidly growing crime typology. NCMEC tracked approximately 100 reports of financial sextortion per day in 2024, and since 2021 is aware of at least 36 teenage boys who took their own lives after victimization [NCMEC, 2024]. In the first half of 2025, CST reports surged from 5,976 to 62,891 [HSToday, 2025].

Investigative burden. The volume of evidence in individual cases has grown exponentially. A single device seizure may contain hundreds of thousands of images, thousands of chat logs, and terabytes of cloud-synced material. The cognitive and psychological load on investigators has scaled accordingly. Digital forensics investigators in child exploitation units experience secondary traumatic stress at rates of 40–60% [Perez et al., 2010, Burns et al., 2008].

Scale of the Problem: 2010 vs. 2024

Metric	≈2010	2024
NCMEC CyberTips per year	∼100,000	29.2M incidents
AI-CSAM CyberTip reports	0	67,000
Online enticement reports	<10,000	546,000
ICAC arrests (annual)	∼5,500	12,600+
Identified victims (cumulative)	<5,000	19,100+

4. Success Rates of Current Enforcement and Online Technology Approaches

4.1. Arrests and Convictions: What the Data Shows

While concrete prosecution rates are difficult to obtain at the federal level, child exploitation cases carry historically high conviction rates. DOJ prosecutions for child sexual exploitation offenses — which include both contact offenses and CSAM-related charges — resolve in conviction in the substantial majority of cases that reach prosecution, reflecting both the weight of digital evidence and the specialized expertise of federal prosecutors. ICAC-referred cases to U.S. Attorneys are typically evidence-heavy: digital forensics, chat logs, device extractions, and hash-matched material provide prosecution teams with evidentiary foundations that are difficult to contest.

State-level data is more variable. The Arkansas ICAC Task Force, over its 20-year history, launched over 9,300 investigations resulting in 1,463 apprehensions — a rough conversion rate of approximately 16% from investigation to arrest, though this figure reflects many other factors including investigation type and available evidence [Arkansas DPS, 2023]. South Carolina's ICAC program arrested and prosecuted hundreds of offenders over a decade, with task force commanders noting that hundreds of individuals who would have victimized children are now incarcerated [SC Attorney General, 2023].

The AZICAC public case reports that CaseLinker was initially evaluated on show 95.7% prosecution outcome coverage — meaning the cases described charges, booking, or sentencing in nearly every instance. This reflects that cases reaching the public reporting stage are those where arrests have occurred. What is harder to track are the cases that did not result in arrest — the investigations that reached dead ends, the tips that could not be acted on due to insufficient information, the jurisdictional handoffs that stalled.

4.2. The Investigation-to-Arrest Gap

The ratio of CyberTip reports to actionable investigations is the most significant disparities in the current system. NCMEC receives tens of millions of reports annually, approximately 1.1 million CyberTips traced to U.S. locations in 2023 and 110,785 reports to the state of California [California ICAC, 2025]. But the fraction of initial reports that result in arrests is a small fraction of the total — not because investigators are failing, but because a substantial portion of tips contain insufficient information to identify the offender and the volume of tips is overwhelming.

NCMEC has flagged this directly: law enforcement has been unable to identify and locate children because online platforms provided inadequate information in their CyberTip reports [Congress, 2025]. The quality of the tip — not just its volume — determines investigative and legal utility.

4.3. Content Detection: Real Limits

Hash-matching technology is the foundation of platform-level CSAM detection, and its effectiveness for known material is well-documented. A reported false positive rate of less than 1 in 1 trillion makes it operationally reliable, and a false negative rate below 2% for known material means very little prior CSAM evades detection on cooperating platforms [Steinebach, 2024, ARES, 2023].

But hash-matching has four structural limitations that become more significant as the threat landscape evolves:

1. **Known-only.** A hash match requires a prior identification. New material — whether generated by AI or captured fresh — will not match any existing hash. This limitation is bounded to perceptual hashing.
2. **Platform-dependent.** Detection only occurs on platforms that have implemented the technology. Small platforms, encrypted messaging services, and dark web networks are outside the detection envelope.
3. **AI adversaries.** Recent cryptographic research has demonstrated that gradient-based attacks can modify an image to avoid PhotoDNA detection while remaining visually similar [Deryck et al., 2026]. As generative AI lowers the barrier to creating new material, the hash-only detection model faces fundamental challenges.

4. **Case blindness.** Hash-matching answers one question: *is this image known illegal material?* It cannot answer any of the investigatively important questions: who created it, how was it distributed, who are the victims, what other cases share common characteristics.

4.4. The Investigator Capacity Problem

At 29.2 million incidents in 2024, the volume of suspected exploitation vastly exceeds the number of investigators available to act on it. This is not a solvable problem through linear scaling of law enforcement resources — the math does not work. At any realistic staffing level, the overwhelming majority of CyberTip reports will never result in an investigation. The response infrastructure requires accurate triage, and triage quality determines which victims get reached.

This is one of the most important and least-discussed aspects of the current approach. The tools that exist — hash-matching, enterprise forensics, case management systems — are good at what they do. But they are not designed to help an investigator decide which of 50 unworked tips represents the most severe active abuse situation. Prioritization at that level requires case-level analysis, not file-level matching.

5. The Importance of Retrospective Case Analysis, AI in Investigations, and CaseLinker

The previous four sections have established the territory. This section makes the affirmative argument: that retrospective case analysis is an underinvested and genuinely important component of the response to child exploitation, that AI integrated carefully into investigative workflows has real potential, and that CaseLinker is one concrete attempt to contribute to both.

5.1. What Retrospective Analysis Can See That Real-Time Detection Cannot

Current detection infrastructure is, by design, real-time. Hash-matching is triggered at the moment of upload. Undercover operations target active predators. Forensic tools examine devices at the moment of seizure. This forward orientation makes sense — active abuse situations require immediate response.

But it creates a systematic blind spot. The patterns that emerge across hundreds or thousands of cases — the platforms that are becoming grooming vectors before they have enough volume to trigger platform-level detection, the perpetrator methodologies that recur across jurisdictions, the demographic risk factors associated with the most severe abuse, the investigation types that correlate with prosecution success — these patterns are only visible in retrospect, in aggregate, and at scale.

> *A single investigator reading one case learns about that case. A system that reads ten thousand cases and surfaces patterns can change how the next investigation is approached, how resources are allocated, and which platforms receive policy attention before a problem becomes a crisis.*

NCMEC and other non-profits publish detailed annual meta-analysis of CyberTipline volume — tracking report counts, ESP breakdown, incident types, and year-over-year trends. That macro-level reporting is valuable and widely cited. What it does not do is systematically analyze the investigative elements of publicly available case reports: the platforms involved, the perpetrator-victim relationship, the offense type, the investigative approach, and how those attributes shift over time and across jurisdictions. That case-level analytical context — derivable from open-source public records — remains largely unaddressed.

Consider a concrete example. At it's 207 cases implementation, the dataset offered only a sliver of the broader investigative landscape — but even that sliver surfaced structure that is invisible at the individual case level. Offense composition became quantifiable: 66.2% of cases involved possession, compared to just 2.9% involving production, with 37.7% involving sexual abuse and 6.8% involving victims under 12. Perpetrator age patterns emerged across cases, with an average documented age gap of 22.6 years between perpetrator and victim — a figure with implications for risk profiling and prevention targeting that no single case could reveal. Agency involvement told its own story: 215 unique law enforcement organizations appeared across the 207 cases, with 79.5% present in only one case — revealing a wide-tailed local network converging around a stable federal core of ICAC task forces, the FBI, and NCMEC. That institutional structure is invisible in any single case report and does not appear in CyberTipline statistics. These are early signals from a small and geographically constrained dataset and is not intended to generalize to the broader landscape of this field. But they illustrate precisely what becomes possible when case-level records are analyzed systematically rather than read in isolation — and what scales with the dataset as it grows. With it's current implementation, 410 cases, and 4 unique sources, agency involvement patterns become even more apparent — which federal partners recur across jurisdictions, which local agencies operate in isolation, and where coordination gaps may exist.

Beyond agency structure, cross-case analysis makes it possible to actively study the recurring differences between distinct case profiles: online-only offenses versus cases involving physical sexual abuse, teacher or authority-figure perpetrators versus strangers, grooming-driven platform contact versus opportunistic exploitation. These profiles carry different risk factors, different investigative requirements, and different policy considerations — but they can only be studied systematically when cases are analyzed in together rather than read individually. This is the value of retrospective analysis: not to replace real-time response, but to inform it.

5.2. The Right Role for AI in This Domain

The conversation about AI in child protection has been dominated by two use cases: AI-generated CSAM (as a new threat) and AI classifiers for CSAM detection (as a defensive tool). Both are important. But there is a third use case that has received less attention: AI for investigative intelligence — using machine learning to extract, organize, and surface patterns from investigative records.

This use case is difficult to execute responsibly. The reasons are clear and were articulated in the second CaseLinker technical report:

1. Case narratives contain sensitive information that cannot be submitted to commercial AI APIs without violating chain of custody and privacy requirements.
2. Outputs from probabilistic ML models cannot be traced to source text and therefore cannot be introduced as evidence.
3. Probabilistic outputs and insights from LLMs cannot be used to reliably cite statistics on the dataset.

These constraints rule out the most obvious and short-cut AI approaches. They do not rule out AI entirely. What they require is that AI be used carefully — as an enhancement to deterministic extraction pipelines rather than a replacement for them, with clear auditability, and with explicit acknowledgment of what each component can and cannot conclude — to aid human investigation into this complex data, not automate it.

The NER integration in CaseLinker's second technical report demonstrates what this looks like in practice. Named entity recognition via Stanza (Stanford NLP) added a semantic extraction layer that identified 215 unique law enforcement organizations across 207 cases — a major increase over what regex alone could capture — while maintaining the core auditability requirement. Every NER extraction can be traced to source text. Pattern-based extraction takes precedence when conflicts arise. Current research explores adding transformers to extract semantic concepts, helping distinguish case topics when regex fails on termonology. The system reverts to its deterministic core if ML components are unavailable or undesired.

This is the goal; responsible AI integration in a sensitive legal context: additive, auditable, optional, and clearly inferior to human judgment for complex interpretive tasks.

5.3. Benefits CaseLinker Has Already Demonstrated

Even at 410 cases — a small dataset — cross-case analysis has surfaced findings that are not visible at the individual case level. These are observations, not conclusions; the dataset is too small for firm statistical claims. But they illustrate the type of intelligence retrospective analysis can generate.

Network structure. 349 unique law enforcement organizations appear across 410 cases. 163 appear in only one case, while federal agencies like ICAC task forces (67.3% of cases), NCMEC (49.7%), and the FBI (7.5%) recur heavily. Federal partners appear less often in raw count but average roughly 8 cases per agency compared to 2 per local agency. This distributed-core structure — a stable federal backbone with a wide-tailed periphery of local partners — is invisible in any single case and has real implications for understanding how ICAC investigations are actually organized and should be supported in practice.

Perpetrator characteristics. Stranger perpetrators account for 95.1% of cases in the expanded dataset, cutting against common assumptions about who commits these crimes. The average age gap between documented perpetrator and victim ages is 20.4 years. Only 2.2% of cases involve registered sex offenders — meaning most perpetrators have no prior exploitation-related convictions.

Offense distribution. Hands-on abuse appears in 32.7% of cases; possession in 78.5%; production in 5.1%; multi-state cases in 3.0%. These distributions, visible in aggregate, are relevant to resource allocation: they can be used to guide resource allocation, prioritize investigative strategies, and align prevention efforts with the most prevalent case types.

Visualizations. Understanding and interacting with this data is challenging. The platform provides 3 separate tabs for interacting with the case material and allows the analysis of a variety of subject matters, including geography, age distribution, severity indicators, and custom topics.

5.4. The Case for Open-Source, Interpretable Tools

One of the deliberate choices in CaseLinker's design is interpretability over algorithmic sophistication and over-engineering. Every extraction can be traced to a source pattern and source text stored in the database. Clustering is performed using weighted Jaccard similarity — a formula an analyst can read, verify, and modify when attempting to compare cases of a similar nature. Priority triage uses a documented scoring function with explicit weights.

Furthermore, the system is entirely modular. If an analyst desires a different visualization or workflow, one can be designed on top of the current database and analysis infrastructure. If an analyst desires specific extracted features or wants to work on their own data, they can modify the Processing pipeline or use the local implementation. The system is designed to aid in analysis of this data, not to dispense pre-packaged conclusions — an analyst who disagrees with a clustering decision, a priority score, or an extracted feature can inspect the source, understand the logic, and change it. The intelligence belongs to the investigator and CaseLinker is a platform that can be use to explore this data.

The open-source release of CaseLinker reflects a similiar commitment: that the analytical capabilities currently available only through expensive enterprise platforms should be accessible to the resource-constrained agencies and nonprofit organizations that do most of the actual child protection work in the United States. A mid-sized ICAC task force with a $1 million annual budget cannot afford Nuix. It can run Python on standard hardware.

5.5. What CaseLinker Is Not

Clarity about limitations matters and there must be limits, especially on technologist.

CaseLinker does not identify victims or offenders. It processes publicly available, already-redacted case summaries — not raw evidence files, not private case databases, not personal data. The patterns it surfaces are patterns in aggregated case summaries, not patterns in individual behavior.

CaseLinker does not replace investigative judgment. As the second technical report notes explicitly: *expert intuition — the ability to recognize meaningful patterns from context and* *experience — remains something algorithms do not yet replicate* [Klein, 1998]. An analyst reading a case understands context, implication, and nuance in ways no NLP or LLM system currently does. CaseLinker is built to support that judgment, not substitute for it.

CaseLinker is not production-ready for active law enforcement use. It is a research platform, developed by me, a graduate student, evaluated on publicly available data, and released as open-source software for further development. The path from research prototype to operational tool involves validation studies, user research with active investigators, and institutional partnerships that do not yet exist.

What CaseLinker is: a demonstration that this type of analysis is possible, useful, and buildable without expensive infrastructure or black-box AI. CaseLinker as it is deployed on GitHub today is a working proof of concept and foundation — one that can be expanded, adapted, and enhanced, while keeping the principles that motivated it.

5.6. Why This Matters Now

Case-level retrospective analysis is more urgent in 2026 than it would have been a decade ago for a specific reason: the threat landscape is changing faster than reporting and detection infrastructure can track.

Ignoring the proliferation of AI and the discussion of agents and their ethical uses, social media and large platform usage has grown substanially over the last decade, shifting the very world that children are growing up in and society is responding to. The scale of connectivity, the normalization of online relationships, and the sheer volume of platforms competing for adolescent attention have all expanded the surface area for exploitation in ways that were not true even in 2015. Sextortion has emerged as a distinct and rapidly scaling crime typology with no clear analog in prior ICAC case history. Online enticement reports increased 192% in 2024 and 77% in the first half of 2025 year-over-year [HSToday, 2025]. The platforms, methods, and perpetrator profiles associated with these new technologies are not yet well-characterized in the public record — precisely because the infrastructure to study them wasn't previously necessary.

Real-time detection is reactive by definition. It catches what has already happened and has been encountered and often prevents abuse for continuing. The analytical value of looking backwards across hundreds of investigated cases complements this; allowing us to understand this ever evolving landscape and map out how technologies have impacted the most vulnerable populations.

> *The goal is not to process cases more efficiently. The goal is to understand and visualize the landscape of child exploitation — to identify emerging platforms, methodologies, and risk factors. That requires looking backwards as carefully as we look forward.*

Conclusion

I started reading ICAC case reports because I learned about this issue from my internet law class. What I found was a problem that is older than the internet, vastly amplified by it, and responded to by a fragmented ecosystem of law enforcement, technology, and policy that is genuinely effective at some things and genuinely overwhelmed by others.

The argument of this piece is narrow and specific: there is an analytical gap at the case level that current tools do not fill, that gap has practical consequences for how resources are allocated and how patterns are recognized, and filling it is technically possible with lightweight, interpretable, open-source tools.

CaseLinker is not a solution to child exploitation. It is a research tool. But tools are important — human beings have used tools to extend what we can see, process, and act on since the beginning of organized problem-solving. The challenge of child exploitation in the digital age, however, is that human judgment alone cannot meet the scale it now operates. At a time when millions of incidents are reported to NCMEC in a single year, where AI is generating new exploitative material, and new exploitative material proliferates across platforms, gaining interpretable insights into the nature of some of the most serious crimes against one of the most vulnerable populations should be a priority.

That is what I am working towards building.

CaseLinker is open-source and available at here. A live demonstration is accessible at here. Report #3 covering 500+ cases, transformer based semantic extraction, and preliminary statistical analysis will be released in April 2026.

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